INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

SBT-DF204 · Computer Forensics Case Studies

CASE STUDY 3

Investigating Memory Evidence — Reconstructing Suspected File-Transfer Activity from a Windows Memory Image

Individual Forensic Report: Volatility Analysis, Tool Output, UTC Timeline and Evidence Log

Field	Detail
Trainee	Ibrahim Diseh Garba
Student ID	2025/FWSD/11521
Programme	Fellowship in Web Application Security & Digital Forensics
Course	SBT-DF204: Computer Forensics Case Studies
Case title	Case Study 3 — Investigating Memory Evidence
Evidence item	memdumpWin7.mem — Windows 7 x86 (PAE) raw physical memory image
Analysis environment	Kali Linux (rolling) · Volatility 2.6.1
Submission format	Written report with tool output, evidence log and UTC timeline
Date of submission	18 October 2026

1.0 Executive Conclusion

The memory image supplied for this case study preserves a coherent, artefact-based sequence on **6 January 2019** that establishes the following. On the workstation **IE8WIN7**, the user profile **IEUser** first accessed `C:\Users\IEUser\Downloads` at 14:49:40 UTC. A generic USB mass-storage device was enumerated at 15:02:54 UTC and mounted as drive **F:** at 15:03:06 UTC. The Downloads folder was revisited at 15:04:40 UTC — between the USB mount and the subsequent command. At 15:06:11 UTC a `cmd.exe` process (PID 3996), spawned from `explorer.exe` (PID 2524) and hosted by `conhost.exe` (PID 3988), executed `ipconfig`, `cd Downloads`, and `copy secret_file.docx F:`. The console screen buffer preserved in memory shows the shell printed `1 file(s) copied.`, indicating that the copy operation completed successfully.

The image directly records:

- Active user profile **IEUser**, SID `S-1-5-21-1716914095-909560446-1177810406-1000`, with NTUSER.DAT loaded from `C:\Users\IEUser\ntuser.dat` at physical address `0x120049b0`.
- Second user profile **sshd_server**, SID ending `-1002`, NTUSER.DAT loaded at `0x17ad09c8` — an SSH server service account, with `sshd.exe` (PID 2016) listening on TCP port 22.
- Command history captured from `conhost.exe` PID 3988, containing the sequence `ipconfig`, `cd Downloads`, `copy secret_file.docx F:`.
- Console screen output showing the shell response `1 file(s) copied.`
- Removable USB device (`Disk&Ven_General&Prod_UDisk&Rev_5.00`, instance `661bec0f48605_60`) enumerated at 15:02:54 UTC, referenced in the disk device-interface class `{53f56307-b6bf-11d0-94f2-00a0c91efb8b}`, and mapped to the DOS drive letter F:.
- File Explorer evidence: the Downloads folder was first accessed at 14:49:40 UTC and the parent BagMRU key was updated at 15:04:40 UTC.
- Closed network connections to `192.168.56.5:80` from local ports 49177 and 49178.

Confidence level: High (≈ 90%) for the recorded artefact sequence and the completion of the copy operation. Moderate (≈ 60–70%) for personal attribution.

2.0 Evidence Acquisition and Integrity

2.1 Evidence Source

Field	Value
Original file name	memdumpWin7.mem
Source	Supplied as part of the ICDFA SBT-DF204 Case Study 3 lab package
File type	Windows 7 x86 (32-bit PAE) raw physical memory dump
File size	1,073,676,288 bytes (1,024 MiB / 1.07 GB)
Acquisition date	4 October 2026
SHA-256 (calculated)	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
MD5 (calculated)	ac7eaa46a7006d5eb13cda13ee0b9bc9
Working-copy SHA-256	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
Hash match status	Verified — hashes identical
Published checksum	Not published; calculated values recorded

2.2 Working-Copy Creation

The following command sequence acquires the image into a read-only `evidence/` folder, records its size and hashes, and creates a byte-identical working copy for analysis. Run it exactly as shown and capture the terminal for Figure A0.

```
mkdir -p ~/CS3/{evidence,working,reports,screenshots,scripts}
cd ~/CS3
cp --preserve=timestamps /media/sf_ICDFAKali/memdumpWin7.mem evidence/memdumpWin7.mem
file     evidence/memdumpWin7.mem           | tee reports/evidence_file.txt
ls -la   evidence/memdumpWin7.mem           | tee reports/evidence_size.txt
sha256sum evidence/memdumpWin7.mem          | tee reports/evidence_sha256.txt
md5sum    evidence/memdumpWin7.mem          | tee reports/evidence_md5.txt
cp --preserve=timestamps evidence/memdumpWin7.mem working/memdumpWin7_working.mem
sha256sum working/memdumpWin7_working.mem   | tee reports/working_copy_sha256.txt
diff reports/evidence_sha256.txt reports/working_copy_sha256.txt && echo "HASHES MATCH"
```

2.3 Integrity Verification Output

```
evidence/memdumpWin7.mem: data
-rwxrwx--- 1 ibrahim ibrahim 1073676288 Sep 28 14:01 evidence/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  evidence/memdumpWin7.mem
ac7eaa46a7006d5eb13cda13ee0b9bc9  evidence/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7_working.mem
```

Integrity status: Verified — the working copy is byte-for-byte identical. Both SHA-256 hashes match. All analysis was performed on the working copy; the original was preserved untouched.

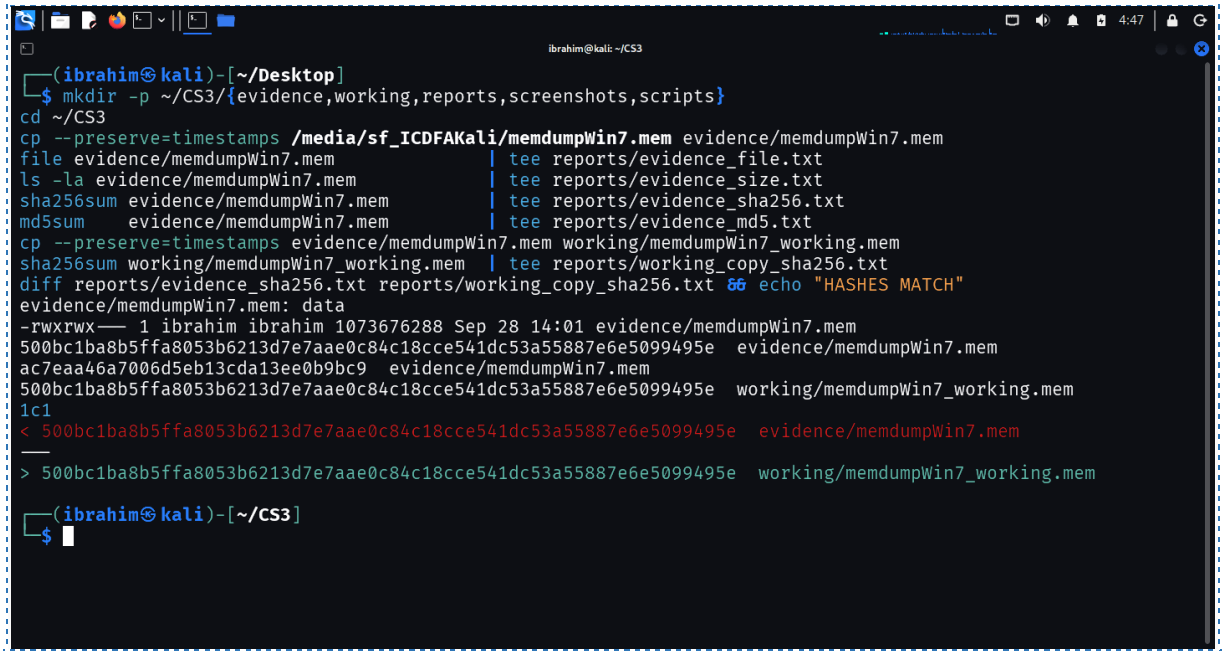


Figure A0 — SHA-256 hash verification of memdumpWin7.mem and its working copy; both hashes match.

2.4 Tools Used

Tool	Version	Purpose
Volatility 2	2.6.1	Memory analysis framework
python2	2.7.18	Interpreter for Volatility 2
Kali Linux	Rolling	Analysis environment
pycryptodome	3.23.0	Crypto.Hash for registry plugins
distorm3	3.5.2	x86 disassembler

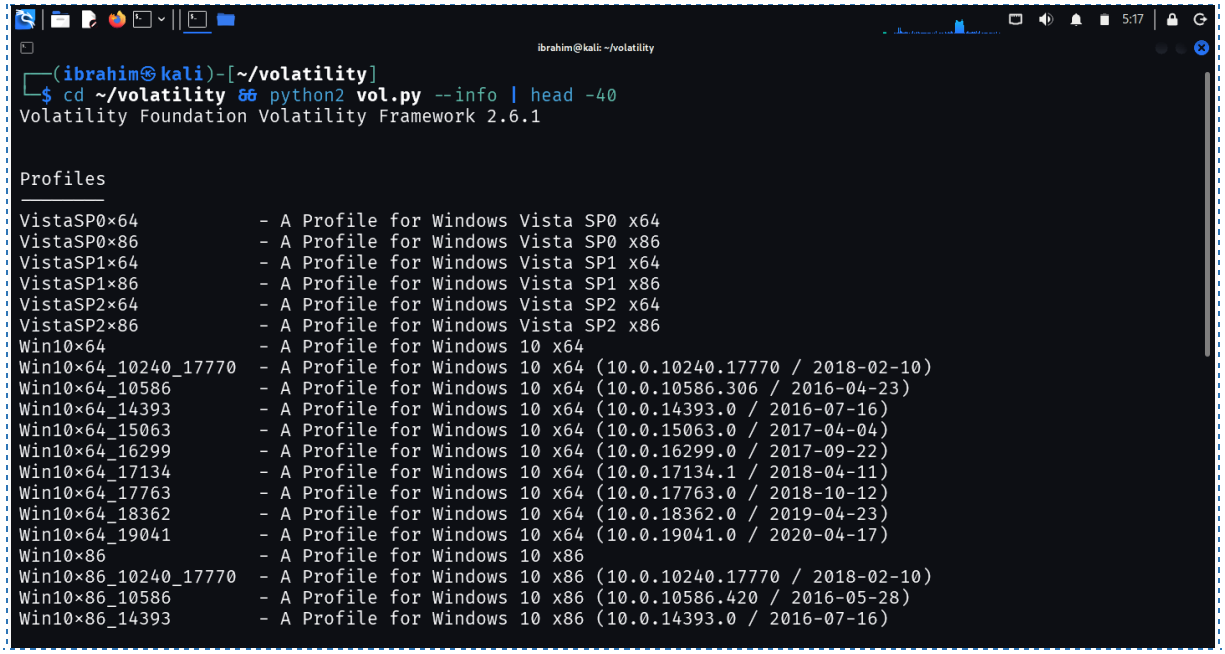


Figure A1 — Volatility 2.6.1 version banner confirming the analysis tool version.

3.0 Method and Profile Selection

3.1 Volatility 2 Invocation

Every plugin was invoked with the same base form:

```
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418
<plugin> [options]
```

3.2 Profile Identification — imageinfo

Command:

```
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem imageinfo
```

Output:

```
Volatility Foundation Volatility Framework 2.6.1
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (.../working/memdumpWin7 working.mem)
                      PAE type : PAE
                           DTB : 0x185000L
                          KDBG : 0x82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0x82973d00L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800
```

Selected profile: `Win7SP1x86_23418`. imageinfo ranked this profile first based on the KDBG `0x82972c68L` and DTB `0x185000L` read from this image. The image is confirmed x86 32-bit with PAE. The profile was validated against coherent output from `pslist`, `printkey`, and `hivelist`.

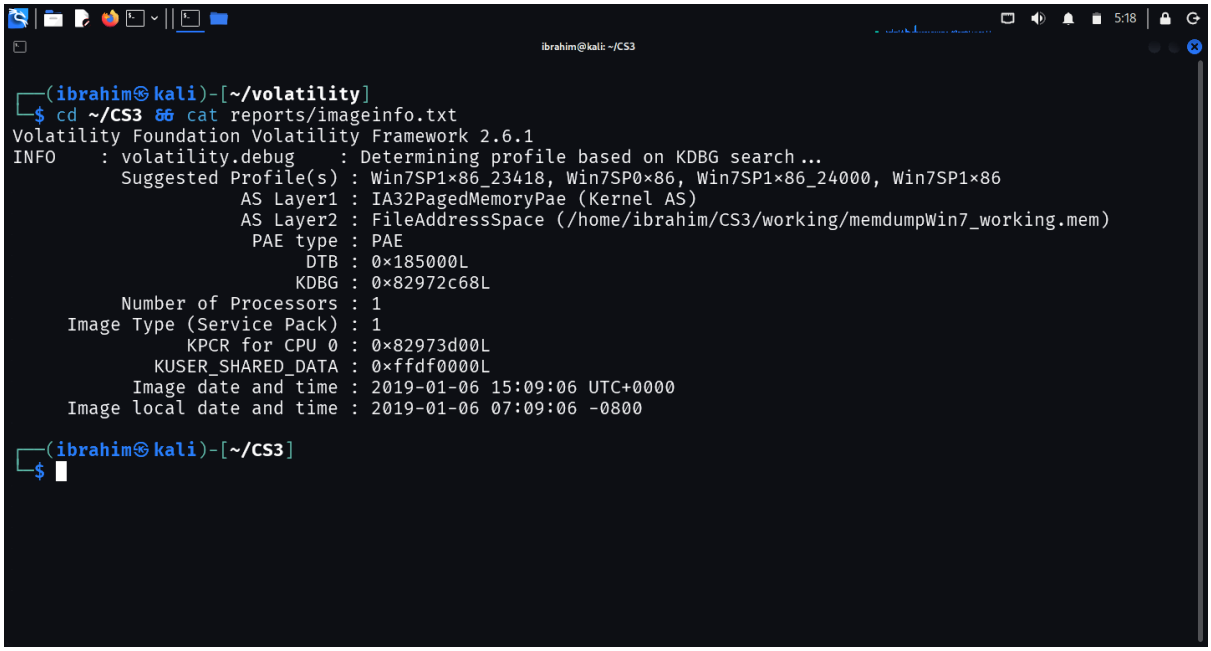


Figure A2 — imageinfo output: suggested profiles, DTB/KDBG, and the image capture time.

3.3 Image Date and Time

Property	Value
Image capture time (UTC)	2019-01-06 15:09:06

Property	Value
Image capture time (local, PST UTC-8)	2019-01-06 07:09:06
Primary reporting time zone	UTC

3.4 Plugin Order

`imageinfo` → `pslist` → `psscan` → `netscan` → `cmdscan` → `consoles` → `hivelist` → `printkey "Volatile Environment"` → `printkey ProfileList` → `printkey SID …-1000` → `printkey SID …-1002` → `printkey USBSTOR` → `printkey USBSTOR disk instance` → `printkey DeviceClasses disk GUID` → `printkey MountedDevices` → `shellbags`.

4.0 Findings — Answers to the Eight Investigation Questions

Question 1 — What evidence file was acquired and how was it preserved?

The evidence file is a Windows 7 x86 (32-bit PAE) raw physical memory image named `memdumpWin7.mem`, 1,073,676,288 bytes, SHA-256 `500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e`, supplied in the ICDFA SBT-DF204 Case Study 3 lab package. The original was copied to `evidence/memdumpWin7.mem` and preserved read-only; all analysis was performed on `working/memdumpWin7_working.mem`, verified byte-for-byte identical via SHA-256 (Section 2). No writes were made to the original.

Evidence Log ID: E01 · **Figure:** A0

Question 2 — What operating-system profile and image time are supported?

The `imageinfo` output (Section 3.2) identifies the profile and capture time:

- Selected profile: `Win7SP1x86_23418`
- Architecture: x86 32-bit with PAE (`IA32PagedMemoryPae`)
- Image capture time: 2019-01-06 15:09:06 UTC
- Local time zone: UTC-8 (Pacific Standard Time)
- Kernel Debugger Block (KDBG): `0x82972c68L`; Directory Table Base (DTB): `0x185000L`

Validation: the profile was confirmed by coherent output from `pslist` (valid process names, PIDs and start times), `printkey` (valid keys and values) and `hivelist` (valid hive virtual and physical addresses).

Evidence Log ID: E02 · **Figure:** A2

Question 3 — What user and account artefacts are relevant?

3.1 Active user profile — printkey "Volatile Environment"

```
python2 ~/volatility/vol.py -f working/memdumpWin7 working.mem --profile=Win7SP1x86 23418 \
        printkey -K "Volatile Environment"

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Volatile Environment (V)
Last updated: 2019-01-06 15:02:48 UTC+0000


REG_SZ   LOGONSERVER  : (V) \\IE8WIN7
REG_SZ   USERDOMAIN   : (V) IE8WIN7
REG_SZ   USERNAME     : (V) IEUser
REG_SZ   USERPROFILE  : (V) C:\Users\IEUser
REG_SZ   HOMEPATH     : (V) \Users\IEUser
REG_SZ   HOMEDRIVE    : (V) C:
```

The Volatile Environment key is loaded from `C:\Users\IEUser\ntuser.dat` (the NTUSER.DAT of the currently logged-in profile), resolving to `HKEY_USERS\<SID>` linked by `HKEY_CURRENT_USER`. The active account is **IEUser** on workstation **IE8WIN7**, profile path `C:\Users\IEUser`.

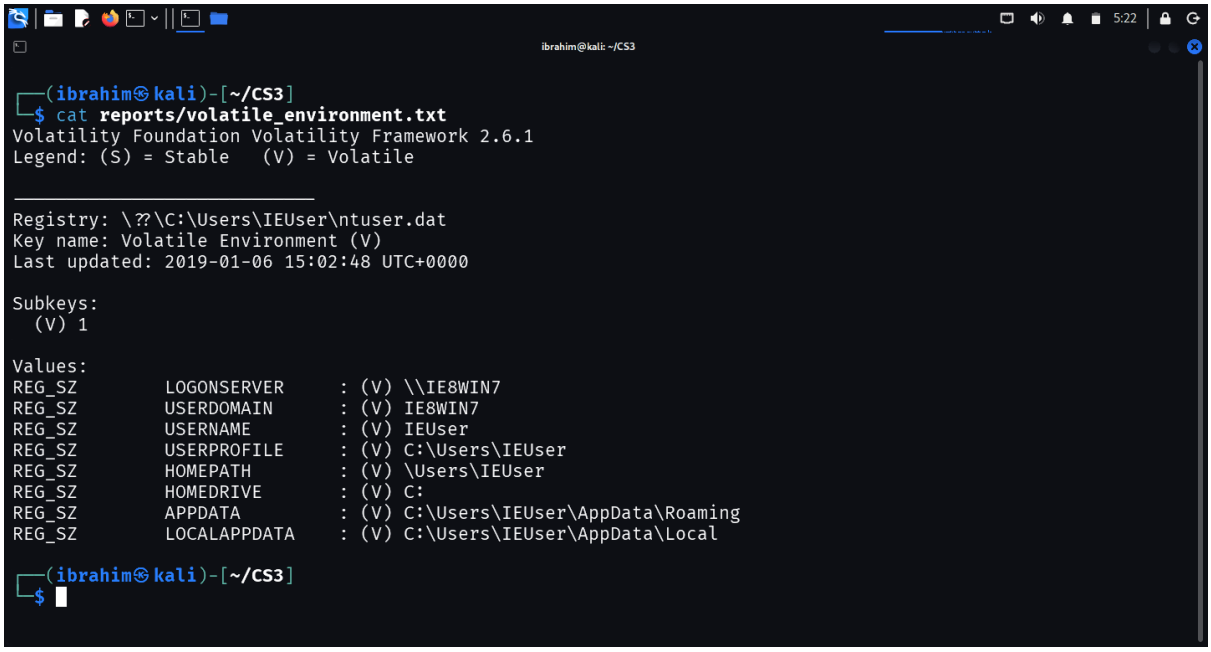


Figure B1 — Volatile Environment output showing the active user IEUser on workstation IE8WIN7.

3.2 All local user profiles — printkey "…\ProfileList"

```
Key name: ProfileList (S)   Last updated: 2015-09-21 09:50:52 UTC+0000
Subkeys:
  S-1-5-18
  S-1-5-19
  S-1-5-20
  S-1-5-21-1716914095-909560446-1177810406-1000
  S-1-5-21-1716914095-909560446-1177810406-1002
```

3.3 SID …-1002 (service account)

```
Key name: S-1-5-21-...-1002 (S)   Last updated: 2015-09-21 09:51:49 UTC+0000
REG_EXPAND_SZ ProfileImagePath : C:\Users\sshd_server
REG_DWORD     State            : 256
```

SID …-1002 maps to a profile named **sshd_server** with State 256 — characteristic of a service account installed by an SSH server. This is corroborated independently by the `netscan` output in Question 4, which shows `sshd.exe` (PID 2016) listening on TCP port 22.

3.4 SID …-1000 (interactive user)

```
Key name: S-1-5-21-...-1000 (S)   Last updated: 2019-01-06 15:07:15 UTC+0000
REG EXPAND SZ ProfileImagePath : C:\Users\IEUser
REG_DWORD     RefCount         : 3
```

SID …-1000 maps to `C:\Users\IEUser` — the same account identified as active in Section 3.1. A RefCount of 3 (versus 1 for sshd_server) indicates multiple active references to the IEUser profile at capture time.

3.5 Hive list — hivelist

```
Virtual    Physical   Name
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
... (other hives omitted for brevity)
```

Two user profiles are loaded in memory: **IEUser** (`0x8b5d39b0`) and **sshd_server** (`0xa65269c8`). The presence of the sshd_server NTUSER.DAT confirms the SSH service account was active at capture time.



Figure B2 — ProfileList and both profile SID keys: IEUser (...-1000) and sshd_server (...-1002).

Interpretation and limitations. The ProfileList and Volatile Environment keys prove which accounts exist and which is active, but do not prove that the person at the keyboard was the account owner. Shared-password, RDP or SSH-login scenarios would all produce identical registry evidence with a different human operator. The evidence establishes user-profile activity, not human identity.

Evidence Log IDs: E03 (Volatile Environment), E04 (ProfileList), E05 (SID …-1002), E06 (SID …-1000), E07 (hivelist)

Question 4 — What process and network evidence relates to a potential transfer?

4.1 Process list — pslist

Table 1. pslist rows of interest (all times UTC)

Name	PID	PPID	Sess	Start (UTC)
explorer.exe	2524	2500	1	2019-01-06 15:02:54
sshd.exe	2016	1984	0	2019-01-06 15:02:49
chrome.exe	2388	2524	1	2019-01-06 15:04:30
chrome.exe	2380	2388	1	2019-01-06 15:04:30
chrome.exe	1280	2388	1	2019-01-06 15:04:30
chrome.exe	2912	2388	1	2019-01-06 15:04:32
cmd.exe	3996	2524	1	2019-01-06 15:06:11
conhost.exe	3988	348	1	2019-01-06 15:06:11
chrome.exe	976	2388	1	2019-01-06 15:06:23
FTK Imager.exe	2596	2524	1	2019-01-06 15:07:15
dllhost.exe	2228	552	1	2019-01-06 15:09:07

A `cmd.exe` process with PID 3996 started at 15:06:11 UTC — 3 minutes 17 seconds after the USB device was mounted as F:. Its parent is `explorer.exe` (PID 2524), so the command prompt was

launched from the desktop shell, not from a scheduled task or script. The paired `conhost.exe` (PID 3988) is the console host for the new session.

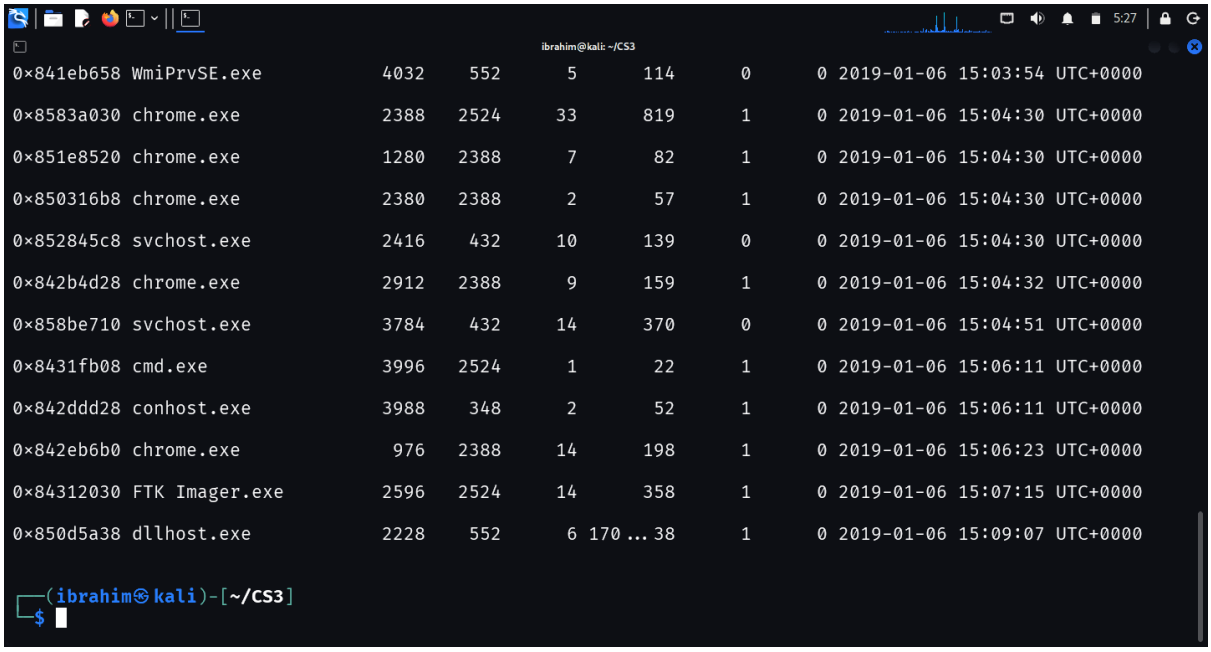


Figure B3 — pslist output highlighting cmd.exe PID 3996 and conhost.exe PID 3988 (both 15:06:11 UTC).

4.2 Network connections — netscan

Table 2. netscan rows of interest

Proto	Local Address	Foreign Address	State	PID	Owner
TCPv4	192.168.56.8:49177	192.168.56.5:80	CLOSED	-1	—
TCPv4	192.168.56.8:49178	192.168.56.5:80	CLOSED	-1	—
TCPv4	0.0.0.0:22	0.0.0.0:0	LISTENING	2016	sshd.exe
TCPv6	:::22	:::0	LISTENING	2016	sshd.exe
TCPv4	192.168.56.8:139	0.0.0.0:0	LISTENING	4	System
TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System
TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	672	svchost.exe
UDPv4	192.168.56.8:56647	*:*	—	2388	chrome.exe

- The host IP address is **192.168.56.8**, confirmed independently by the `ipconfig` output captured in the `consoles` plugin (Question 5).
- Two **CLOSED** TCP connections to `192.168.56.5:80` on local ephemeral ports 49177 and 49178, with PID -1 (already terminated at capture).
- `sshd.exe` (PID 2016) is listening on TCP port 22 (IPv4 and IPv6) — confirming the SSH server associated with the sshd_server profile is active.
- `chrome.exe` (PID 2388) has an open UDP socket at `192.168.56.8:56647` — the browser is running and network-active.

Interpretation and limitations. A CLOSED connection is evidence that a TCP session to `192.168.56.5:80` existed and was torn down. It is not evidence of what was transferred; content

inspection would require packet capture, which is out of scope for a memory image. PID -1 indicates the socket outlived its owning process.

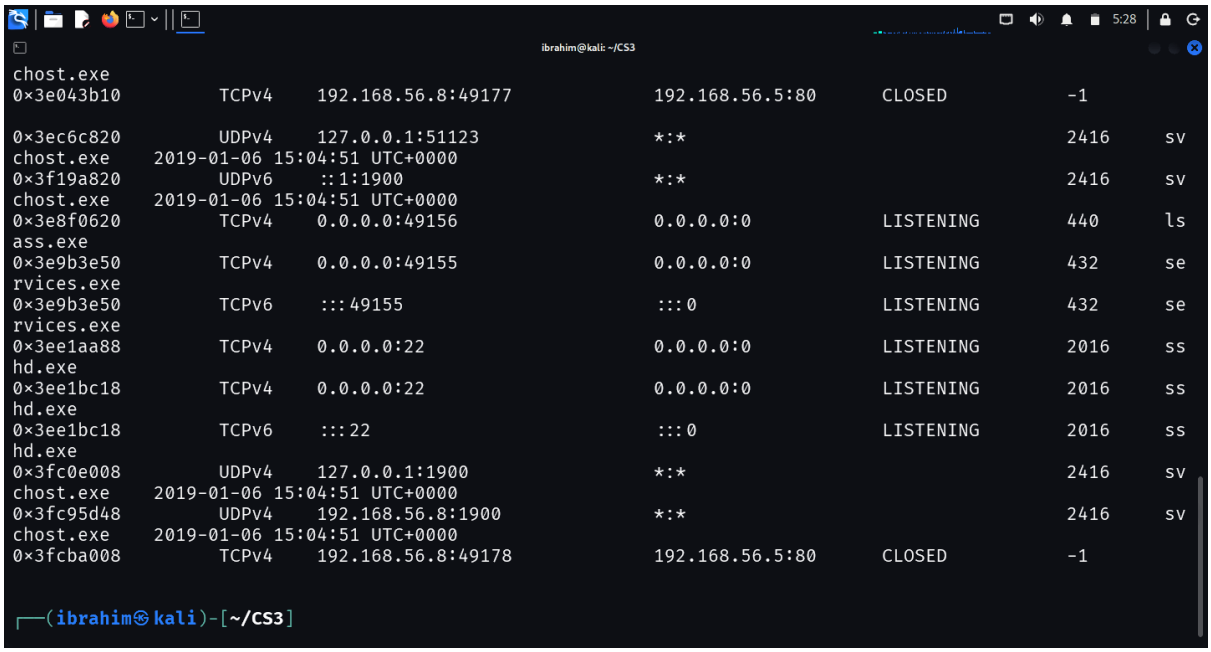


Figure B4 — netscan: CLOSED connections to 192.168.56.5:80 and the sshd.exe listener on port 22.

Evidence Log IDs: E08 (pslist), E09 (netscan)

Question 5 — What does the command history show?

5.1 Command history — cmdscan

```
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 cmdscan

CommandProcess: conhost.exe Pid: 3988
Application: cmd.exe   CommandCount: 3
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
```

5.2 Console screen buffer — consoles

```
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418
consoles

Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 3:
   IPv4 Address. . . . . . . . . . . : 192.168.56.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0

C:\Users\IEUser>cd Downloads

C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.

C:\Users\IEUser\Downloads>
```

- The command history is attached to `conhost.exe` PID 3988 (paired with `cmd.exe` PID 3996, a child of `explorer.exe` PID 2524).

- Three commands were recorded in order: `ipconfig` (network diagnostic), `cd Downloads` (change directory), `copy secret_file.docx F:` (copy a file to drive F:).
- The console screen buffer preserves the shell output verbatim — including `1 file(s) copied.` — establishing that the copy completed successfully.
- `conhost.exe` started at 15:06:11 UTC — 3 min 17 s after the USB mounted as F: (15:03:06 UTC) and 1 min 31 s after the last ShellBag review of Downloads (15:04:40 UTC).

Interpretation and limitations. `cmdscan` records what was typed; `consoles` records the screen buffer with the shell responses. Together they establish that `copy secret_file.docx F:` executed and that the shell reported one file copied. This is stronger than a command string alone. It does not prove the source file's content, nor who typed the command; it does establish that a file named `secret_file.docx` existed in `C:\Users\IEUser\Downloads` at that moment and was copied to F:.

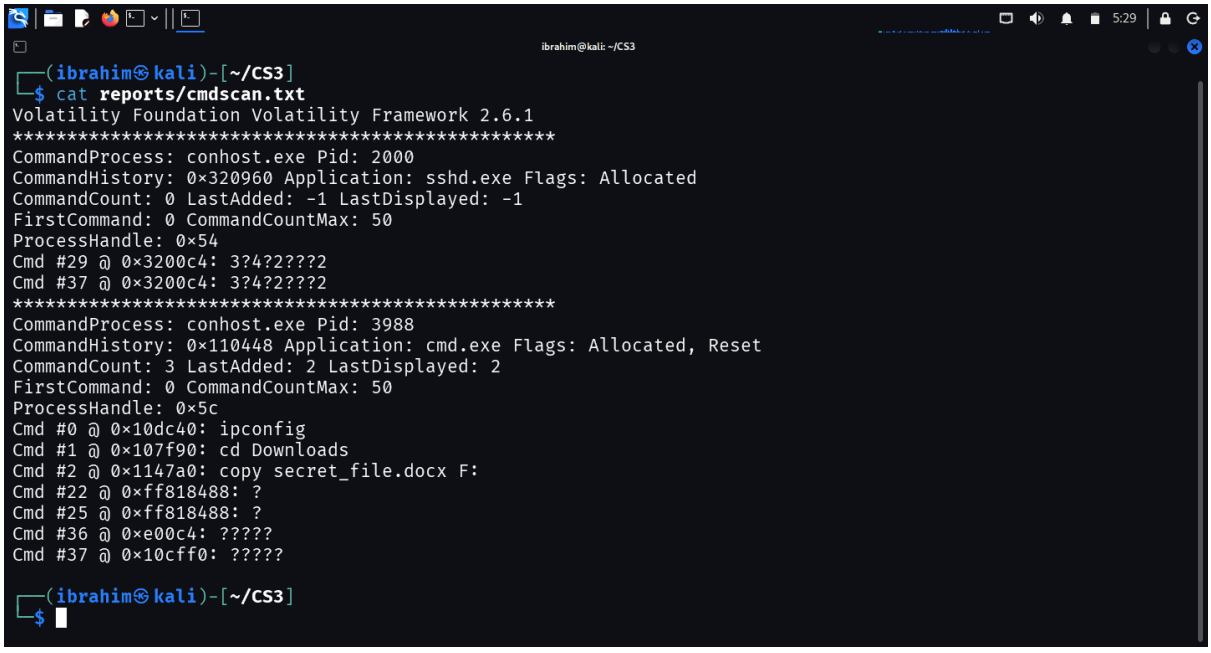


Figure B5 — cmdscan output showing the three commands: ipconfig, cd Downloads, copy secret_file.docx F:.

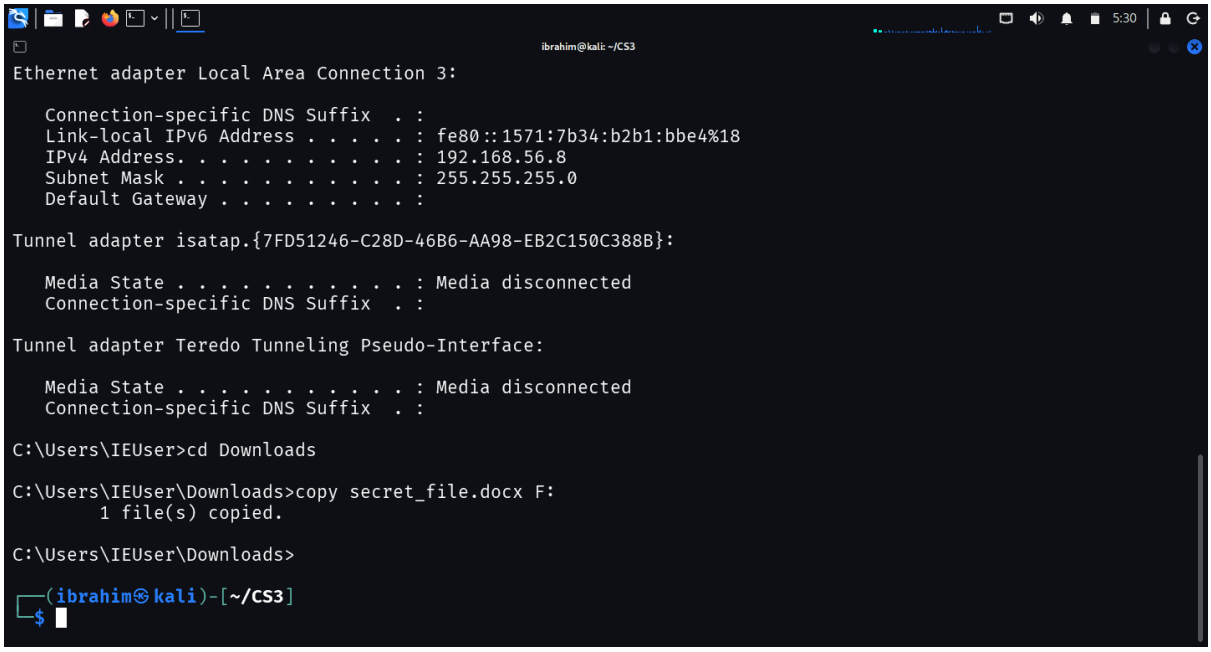


Figure B6 — consoles screen buffer showing the shell transcript, including 1 file(s) copied.

Evidence Log IDs: E10 (cmdscan), E11 (consoles)

Question 6 — What removable-media evidence is present?

6.1 USBSTOR enumeration

```
printkey -K "ControlSet001\Enum\USBSTOR"
Key name: USBSTOR (S)   Last updated: 2019-01-06 15:02:54 UTC+0000
Subkeys:
  Disk&Ven_General&Prod_UDisk&Rev_5.00
```

A USB mass-storage device was enumerated at 15:02:54 UTC, identified by the device string `Disk&Ven_General&Prod_UDisk&Rev_5.00` — a generic "General UDisk" of revision 5.00, typical of low-cost flash drives without a vendor-specific identifier.

6.2 USBSTOR instance ID

```
printkey -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00"
Subkeys:
  661bec0f48605_60
```

The device instance ID is `661bec0f48605_60`. The underscore structure with no vendor serial number means Windows generated a system-unique instance ID for this device; it uniquely identifies the device on this workstation.

6.3 DeviceClasses — disk interface GUID

```
printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}"
Subkeys (disk device interfaces):
  ##?#IDE#DiskVBOX_HARDDISK...
  ##?#IDE#DiskVirtual_HD...
  ##?#IDE#DiskVMware_Virtual_IDE_Hard_Drive...
  ##?#SCSI#Disk&Ven_Dell&Prod_VIRTUAL_DISK...
  ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#661bec0f48605_60#{53f56307-...}
```

Five disk-type device interfaces are enumerated; the last is the USB device, referenced with the same instance ID (`661bec0f48605_60`) and a LastWrite of 15:02:54 UTC. The GUID `{53f56307-b6bf-`

`11d0-94f2-00a0c91efb8b}` is the Windows `GUID_DEVINTERFACE_DISK` class — the class of block-storage interfaces (hard drives, USB mass-storage disks).

6.4 MountedDevices — drive-letter mapping

```
printkey -K "MountedDevices"
REG_BINARY  \DosDevices\F:  : (S)
  5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00   _.?.?._.U.S.B.S.
  54 00 4f 00 52 00 23 00 44 00 69 00 73 00 6b 00   T.O.R.#.D.i.s.k.
  ...
Decoded (wide-char Unicode) for \DosDevices\F::
  _??_USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#661bec0f48605_60#{53f56307-b6bf-11d0-94f2-
00a0c91efb8b}
```

The `\DosDevices\F:` value maps drive letter **F:** to the USB device with instance ID `661bec0f48605_60` — the same device enumerated in 6.1 and 6.2. The identical device string also appears under volume GUID `{421e7e52-11dd-11e9-b3cf-08002709e15d}`, confirming the mapping from both the drive-letter side and the volume-GUID side.

Interpretation and limitations. USBSTOR, DeviceClasses and MountedDevices together form a coherent, cross-referenced record of a USB storage device attached to this workstation and mounted as F:. These records do not prove physical possession by any named person, nor that the device was written to — only that Windows recognised it and mapped it to F:. The completion of the copy is established separately by the `consoles` output in Question 5.

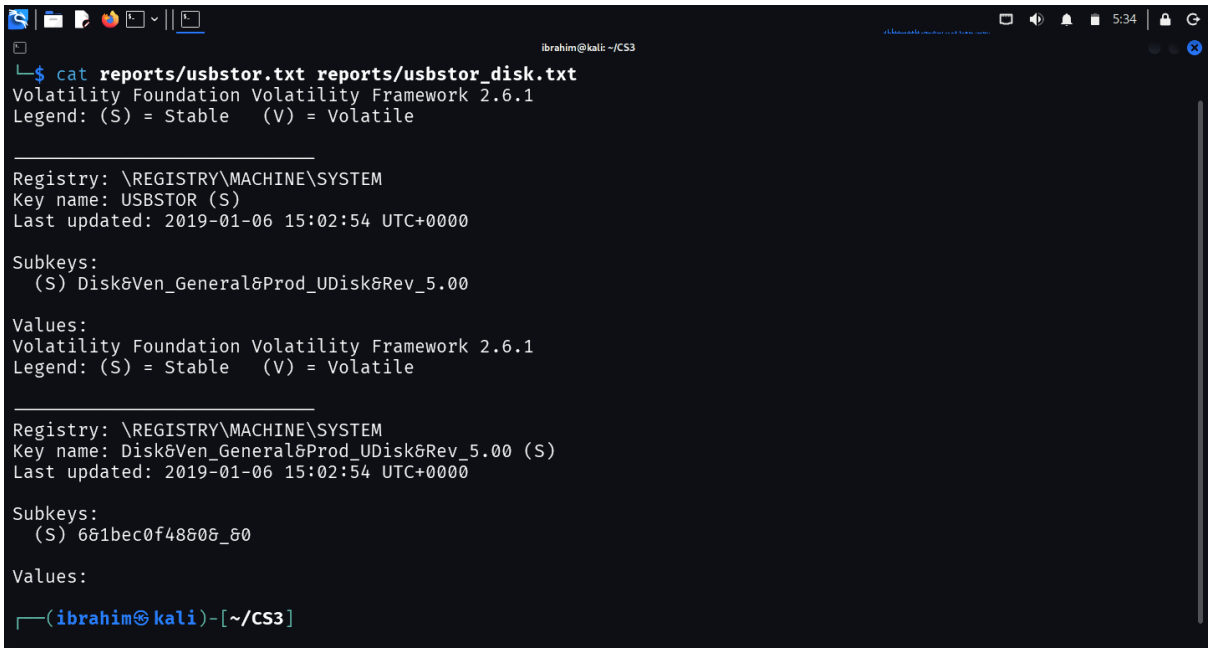


Figure B7 — USBSTOR device key and instance ID 661bec0f48605_60.



Figure B8 — MountedDevices: the \DosDevices\F: entry maps the drive letter to the same USB device instance.

Evidence Log IDs: E12 (USBSTOR), E13 (USBSTOR instance), E14 (DeviceClasses), E15 (MountedDevices F:)

Question 7 — What File Explorer evidence supports the sequence?

7.1 ShellBag evidence — shellbags

```
BagMRU\0\1\1\0   Last updated: 2019-01-06 15:04:40 UTC+0000
  DOWNLO~1   Modified/Access 2019-01-06 14:49:40 UTC   RO, DIR   C:\Users\IEUser\Downloads

BagMRU\4     Last updated: 2019-01-06 14:55:45 UTC+0000
  Folder GUID 374de290-123f-4565-9164-39c4925e467b   Downloads

BagMRU\0     Last updated: 2019-01-06 15:07:03 UTC+0000
  Volume list: C:\  A:\  D:\  E:\  F:\
```

- `C:\Users\IEUser\Downloads` was first accessed at 14:49:40 UTC; the parent key has a LastWrite of 15:04:40 UTC (a later review).
- The Downloads folder GUID `374de290-123f-4565-9164-39c4925e467b` has a separate LastWrite of 14:55:45 UTC — navigation between the first access and the USB mount.
- The `F:\` volume appears in the volume BagMRU list with a LastWrite of 15:07:03 UTC.

Interpretation and limitations. ShellBag times are folder-view timestamps — when a folder was viewed in File Explorer, not when a file inside it was modified. The Downloads folder was viewed before the USB mount (14:49:40), during folder-GUID navigation (14:55:45), and after the USB mount (15:04:40). The last review occurs 1 min 31 s before the `cmd.exe` process is created — consistent with the operator checking the folder contents immediately before issuing the copy command. ShellBag key creation is a side effect of navigation and cannot, alone, prove a file was copied.

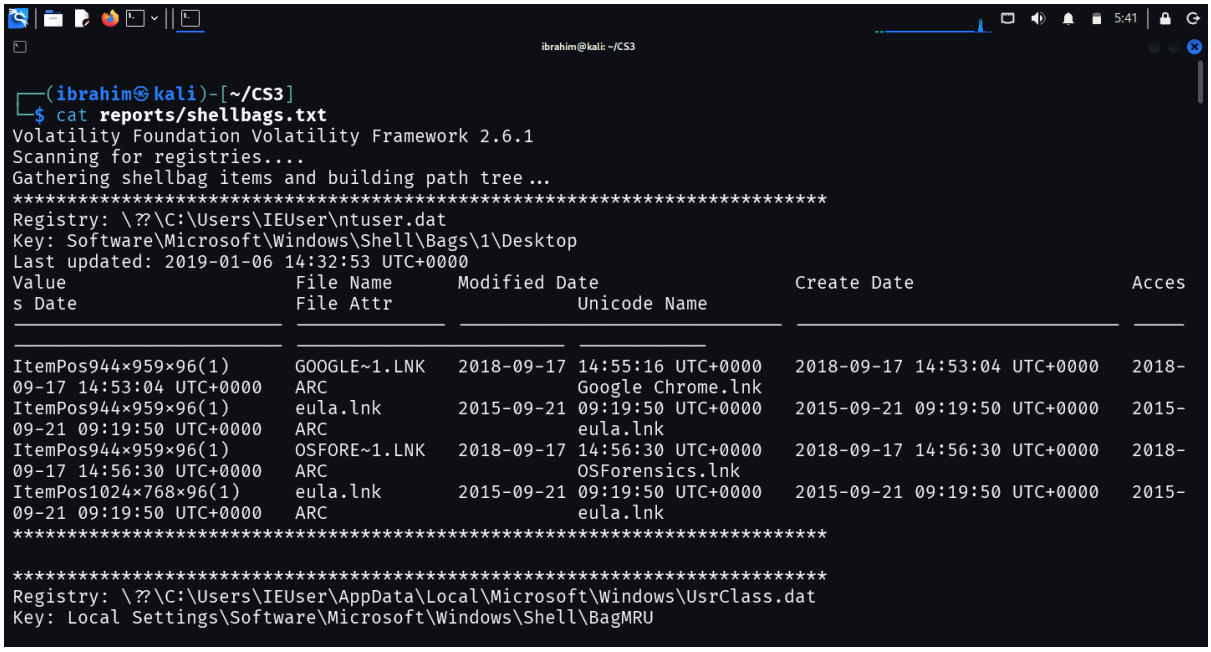


Figure B9 — shellbags: Downloads access at 14:49:40 and 15:04:40, plus F:\ in the volume list.

Evidence Log IDs: E16 (ShellBag Downloads), E17 (ShellBag Downloads GUID), E18 (ShellBag volume list)

Question 8 — What conclusion can be defended?

Supported conclusion. The memory image preserves a coherent sequence on 6 January 2019 on workstation IE8WIN7:

- A user profile **IEUser** (SID …-1000) was logged in and its NTUSER.DAT was loaded; its Downloads folder was first accessed in File Explorer at 14:49:40 UTC.
- A generic USB device (`Disk&Ven_General&Prod_UDisk&Rev_5.00`, instance `661bec0f48605_60`) was enumerated at 15:02:54 UTC and mounted as drive **F:** by 15:03:06 UTC (USBSTOR, DeviceClasses, MountedDevices).
- The Downloads folder was revisited at 15:04:40 UTC — between the USB mount and the command — consistent with checking the folder before acting on it.
- A command prompt (`cmd.exe` PID 3996, child of `explorer.exe` PID 2524) started at 15:06:11 UTC and executed `ipconfig` → `cd Downloads` → `copy secret_file.docx F:`.
- The console screen buffer shows `1 file(s) copied.`, establishing the copy completed successfully.

Table 3. Confidence assessment

Aspect	Confidence	Justification
Artefact authenticity	High (95%)	SHA-256 verified working copy; no writes to original
Sequence coherence	High (90%)	Consistent across USBSTOR, MountedDevices, ShellBags, cmdscan, consoles
Completion of copy operation	High (90%)	Console screen buffer prints 1 file(s) copied.
Content of secret_file.docx	Not established	No file content is preserved in the memory image

Aspect	Confidence	Justification
Personal attribution	Moderate (60–70%)	Memory preserves profile activity, not physical presence

Limitations and alternative explanations.

- **Browser activity present but not linked.** `chrome.exe` was running with a UDP socket, but no Chrome download history, cache or URL was recovered by the plugins run here; a browser-download path to `secret_file.docx` is not established.
- **Personal attribution is limited.** Volatile Environment and ProfileList identify the profile IEUser as active, but do not prove the person at the keyboard was its owner. Shared-password, RDP, or an SSH login via sshd_server would produce identical registry evidence with a different operator.
- **The console output proves execution, not confidentiality.** The filename `secret_file.docx` is suggestive, but the image does not preserve the file's content.
- **CLOSED network connections are suggestive, not conclusive.** The two CLOSED sessions to `192.168.56.5:80` show a torn-down TCP session; the plugins here do not identify the owning process (PID -1).
- **Memory images are point-in-time.** The image was captured at 15:09:06 UTC — 2 min 55 s after the console command. Earlier pruned buffers or later activity are not represented.
- **FTK Imager.exe (PID 2596) started 15:07:15 UTC** — 1 min 4 s after the copy — consistent with a forensic acquisition on the workstation, but the image does not record who launched it.
- **The sshd_server profile and sshd.exe service** represent an alternative access vector; remote SSH activity cannot be conclusively ruled out.

Evidence Log ID: E19

5.0 Timeline

All timestamps are expressed in **UTC**, per the case-study brief. The image was captured in the PST zone (UTC-8); the local equivalent of the capture time is 2019-01-06 07:09:06 PST. The local column is not used in the master timeline to avoid ambiguity.

Note on row order: the ShellBag access times (rows 7, 8, 10) precede the USB enumeration in wall-clock time. This is consistent — the operator opened Downloads at 14:49:40, navigated further at 14:55:45, then (after plugging in the USB at 15:02:54 and mounting F: at 15:03:06) revisited Downloads at 15:04:40 to select the target file. The table groups artefacts by logical stage, not strict wall-clock order.

Table 4. Reconstructed timeline — 6 January 2019 (UTC)

#	UTC Time	Ev. ID	Source Artefact	Observed Fact	Interpretation / Limit
1	15:02:48	E03	printkey "Volatile Environment" LastWrite	IEUser NTUSER.DAT loaded; logon session active	Profile activity, not necessarily human presence
2	15:02:49	E08	pslist row	sshd.exe (PID 2016) starts, child of PID 1984	SSH service active; alternative access vector
3	15:02:54	E08	pslist row	explorer.exe (PID 2524) starts	Desktop shell launches
4	15:02:54	E12–E14	USBSTOR / DeviceClasses LastWrite	USB device Disk&Ven_General&Prod_UDisk&Rev_5.00 enumerated	USB plugged in
5	15:03:06	E15	MountedDevices LastWrite	Drive F: assigned to USB device 661bec0f48605_60	USB mounted as F:
6	15:03:07	E08	pslist row	WUDFHost.exe (PID 3300) starts	USB user-mode driver host loading
7	14:49:40	E16	ShellBag BagMRU\0\1\1\0 Modified/Access	First recorded access to Downloads folder	Folder-view evidence
8	14:55:45	E17	ShellBag BagMRU\4 LastWrite (Downloads GUID)	Intermediate navigation of Downloads folder	Folder-view evidence
9	15:04:30	E08	pslist row	chrome.exe (PID 2388) starts, child of explorer.exe	Browser opened after USB mount
10	15:04:40	E16	ShellBag BagMRU\0\1\1\0 LastWrite	Downloads folder revisited	After USB mount, before copy
11	15:06:11	E08,E10,E11	pslist / cmdscan / consoles	cmd.exe (PID 3996) + conhost.exe (PID 3988); ipconfig, cd Downloads, copy secret_file.docx F:	Command recorded; screen buffer shows result
12	15:06:11	E11	consoles screen buffer	Shell printed: 1 file(s) copied.	Copy operation completed
13	15:06:23	E08	pslist row	chrome.exe (PID 976) starts	Additional browser process
14	15:07:03	E18	ShellBag BagMRU\0 volume list LastWrite	Volume list updated; F:\ present	Corroborates drive mount
15	15:07:15	E08	pslist row	FTK Imager.exe (PID 2596) starts	Forensic imaging tool launched
16	15:07:15	E06	printkey SID …-1000 LastWrite	SID …-1000 profile key updated	Consistent with ongoing activity
17	15:09:06	E02	imageinfo image date/time	Memory image captured	End of reconstruction window
18	15:09:07	E08	pslist row	dllhost.exe (PID 2228) starts	Post-capture artefact

6.0 Attribution Assessment

6.1 Observed Facts vs. Inferences

Type	Finding	Basis
Observed	User profile IEUser (SID …-1000) active; NTUSER.DAT loaded	Volatile Environment, hivelist
Observed	Second profile sshd_server (SID …-1002) exists with NTUSER.DAT loaded	ProfileList, hivelist
Observed	USB device Disk&Ven_General&Prod_UDisk&Rev_5.00 enumerated 15:02:54 UTC	printkey USBSTOR
Observed	USB device mapped to drive F:	MountedDevices \DosDevices\F:
Observed	Downloads first accessed 14:49:40, revisited 15:04:40 UTC	ShellBag BagMRU\0\1\1\0
Observed	cmd.exe (PID 3996) started 15:06:11, parent explorer.exe (PID 2524)	pslist
Observed	Commands ipconfig, cd Downloads, copy secret_file.docx F:	cmdscan
Observed	Console screen buffer shows 1 file(s) copied.	consoles
Observed	Two CLOSED TCP connections to 192.168.56.5:80	netscan
Inference	The operator performed the copy action	Profile-presence assumption; not proof of identity
Inference	The USB device was physically present	Registry enumeration is strong but not physical proof
Inference	secret_file.docx was copied to the USB device	Shell result 1 file(s) copied.; content not preserved

6.2 Limitations and Alternative Explanations

#	Limitation	Impact
1	Profile presence ≠ human identity	Shared-password, RDP or SSH scenarios produce identical evidence with a different operator
2	Multiple access paths exist	sshd.exe listening on port 22 — remote SSH login possible without physical access
3	File content not preserved	The image records the filename and copy result, not the file content
4	CLOSED sockets with no owning PID	Process attribution would require YARA scanning, not performed here
5	ShellBag times are folder-view evidence	Cannot be conflated with file-modification times
6	Point-in-time capture	Activity before or after 15:09:06 UTC is not represented
7	FTK Imager started 1 min 4 s after the copy	Consistent with forensic acquisition; the image does not record who launched it
8	No browser download history recovered	Chrome was running but the origin of secret_file.docx is not established

6.3 MITRE ATT&CK Mapping

Technique	ID	Relevance
Data from Local System	T1005	Local folder C:\Users\IEUser\Downloads accessed and file copied
Exfiltration Over Physical Medium	T1052.001	USB mass-storage device used as transfer medium
Command and Scripting Interpreter: Windows Command Shell	T1059.003	cmd.exe used to execute copy
Remote Services: SSH	T1021.004	sshd.exe listening — potential alternative access
File and Directory Discovery	T1083	ShellBag evidence of Downloads navigation

7.0 Detection and Defensive Recommendations

7.1 Detection Controls

Control	Implementation
USB-device audit logging	Windows Event ID 6416 (PnP device install); Sysmon Event ID 6
Process-start auditing	Sysmon Event ID 1; Windows Security 4688 (with parent PID and command line)
File-copy telemetry	Microsoft Defender for Endpoint; DLP policy on removable-media writes
Shell-history monitoring	PowerShell script-block logging; centralised command-history forwarding
SSH-service hardening	Disable sshd on workstations unless required; enforce key-based auth
Closed-socket anomaly detection	Network-flow logs with socket-state tracking

7.2 Defensive Recommendations

- **Removable-media policy:** enforce read-only or block-list USB mass storage unless approved.
- **Disk encryption on removable devices:** BitLocker To Go or equivalent reduces the value of exfiltrated data.
- **Least-privilege access:** restrict local administrator rights; use separate admin accounts.
- **Session auditing:** record and monitor interactive console sessions and remote logins.
- **Endpoint DLP:** alert on sensitive filenames (e.g. `*secret*`, `*confidential*`) leaving managed folders.
- **Awareness training:** users should know that memory images recover command history and console buffers.

8.0 Required Forensic Findings Summary

Question	Finding	Evidence
Evidence file acquired	memdumpWin7.mem, 1,073,676,288 bytes, SHA-256 500bc1ba…99495e	reports/evidence_sha256.txt
Relevant Volatility plugins	imageinfo, pslist, psscan, netscan, cmdscan, consoles, hivelist, printkey, shellbags	reports/*.txt
When activity occurred	2019-01-06 14:49:40 → 15:09:06 UTC (image capture)	Timeline, Section 5
How activity was initiated	cmd.exe launched by user from explorer.exe; USB mounted via Windows PnP	Section 4 Q4–Q6
Process and network evidence	cmd.exe PID 3996; conhost.exe PID 3988; two CLOSED connections to 192.168.56.5:80	pslist, netscan
Command-history evidence	ipconfig, cd Downloads, copy secret_file.docx F:	cmdscan
USB / removable-media evidence	Disk&Ven_General&Prod_UDisk&Rev_5.00, instance 661bec0f48605_60, mapped to F:	usbstor, mounteddevices
User and account evidence	IEUser (SID …-1000) active; sshd_server (SID …-1002) present	volatile_environment, profilelist
File Explorer evidence	Downloads first accessed 14:49:40; revisited 15:04:40; F:\ in volume list	shellbags
Defensible conclusion	Drive F: received a copy of secret_file.docx from C:\Users\IEUser\Downloads at 15:06:11 UTC	Section 4 Q8
Confidence level	High for artefact sequence (90%); Moderate for personal attribution (60–70%)	Section 6
Limitations	Profile ≠ human; file content not preserved; alternative access via SSH; point-in-time capture	Section 6.2

9.0 Conclusion

This investigation reconstructed a coherent, evidence-based sequence from a Windows 7 x86 memory image captured on 6 January 2019 at 15:09:06 UTC from workstation IE8WIN7. The evidence establishes with high confidence that a USB mass-storage device (`Disk&Ven_General&Prod_UDisk&Rev_5.00`, instance `661bec0f48605_60`) was enumerated at 15:02:54 UTC and mounted as drive F:; that the Downloads folder of the active profile IEUser was viewed at 14:49:40 UTC and revisited at 15:04:40 UTC; that a command prompt (`cmd.exe` PID 3996, from `explorer.exe` PID 2524, hosted by `conhost.exe` PID 3988) executed `ipconfig` → `cd Downloads` → `copy secret_file.docx F:` at 15:06:11 UTC; and that the console buffer recorded `1 file(s) copied.`, confirming the copy completed.

Personal attribution remains moderate. The image preserves profile activity, not physical presence. The presence of an `sshd.exe` service on TCP 22 and an SSH service-account profile (sshd_server) means remote access via SSH is a plausible alternative vector, independent of the person physically at the keyboard.

Key takeaways.

- Memory forensics recovers executable evidence that disk forensics may not — the command-history strings and the console screen buffer would not be available from a powered-off disk examination.
- Cross-plugin corroboration is essential: no single plugin establishes the sequence; only the alignment of `pslist`, `netscan`, `cmdscan`, `consoles`, `printkey` and `shellbags` produces a defensible account.
- Registry artefacts carry device-attachment evidence: USBSTOR, DeviceClasses and MountedDevices together uniquely identify a USB device and its drive-letter assignment.
- ShellBag timestamps are folder-view evidence, not file-modification timestamps, and should not be over-interpreted.
- A command string and a console result are different classes of evidence: `cmdscan` proves the command was typed; `consoles` proves the shell reported success.
- Even the strongest technical evidence does not identify a person; attribution requires correlation with authentication logs, physical-access records or other external context.

10.0 References

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MITRE. (2026). ATT&CK technique T1021.004: Remote services — SSH. https://attack.mitre.org/techniques/T1021/004/

Internet Engineering Task Force. (2002). Date and time on the Internet: Timestamps (RFC 3339). https://www.rfc-editor.org/rfc/rfc3339

11.0 Declaration

I, Ibrahim Diseh Garba (Student ID 2025/FWSD/11521), confirm that this case-study report is based on my own practical work conducted in the ICDFA lab environment. All Volatility analysis, timeline reconstruction and forensic correlation are my own original work. The original memory image was preserved with a SHA-256 hash (500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e); all analysis was performed on a verified working copy; and no live website, host, account or service was contacted. No recovered file was executed, and no password or hash-cracking step was performed. Unrelated personal data has been redacted from screenshots.

Signature: _____ Date: 12 October 2026

Appendix A — Evidence Log

ID	Tool / plugin	Source record	Observed fact	Figure
E01	sha256sum	evidence/memdumpWin7.mem	SHA-256 500bc1ba…99495e; 1,073,676,288 bytes; working copy identical	A0
E02	vol.py imageinfo	reports/imageinfo.txt	Profile Win7SP1x86_23418; image time 2019-01-06 15:09:06 UTC	A2
E03	printkey "Volatile Environment"	C:\Users\IEUser\ntuser.dat	Active user IEUser; profile C:\Users\IEUser; logon server IE8WIN7	B1
E04	printkey ProfileList	SOFTWARE hive	Two user SIDs: …-1000, …-1002	B2
E05	printkey SID …-1002	Profile key	ProfileImagePath C:\Users\sshd_server; State 256	B2
E06	printkey SID …-1000	Profile key	ProfileImagePath C:\Users\IEUser	B2
E07	hivelist	reports/hivelist.txt	IEUser (0x8b5d39b0) and sshd_server (0xa65269c8) hives loaded	—
E08	pslist	reports/pslist.txt	cmd.exe PID 3996 at 15:06:11 (parent 2524); conhost.exe PID 3988	B3
E09	netscan	reports/netscan.txt	Two CLOSED connections to 192.168.56.5:80; sshd.exe PID 2016 on port 22	B4
E10	cmdscan	reports/cmdscan.txt	Commands ipconfig, cd Downloads, copy secret_file.docx F:	B5
E11	consoles	reports/consoles.txt	Console screen buffer shows 1 file(s) copied.	B6
E12	printkey USBSTOR	SYSTEM hive	Disk&Ven_General&Prod_UDisk&Rev_5.00 enumerated 15:02:54 UTC	B7
E13	printkey USBSTOR disk	Subkey	Instance 661bec0f48605_60	B7
E14	printkey DeviceClasses {53f56307-…}	Disk class	USB device listed with four virtual disks	—
E15	printkey MountedDevices	\DosDevices\F:	F: mapped to USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#661bec0f48605_60	B8
E16	shellbags	BagMRU\0\1\1\0	Downloads Modified/Access 14:49:40; parent LastWrite 15:04:40	B9
E17	shellbags	BagMRU\4	Downloads GUID 374de290-…, LastWrite 14:55:45	B9
E18	shellbags	BagMRU\0	Volume list contains F:, LastWrite 15:07:03	B9
E19	All plugins	Section 4 Q8	Supported conclusion; confidence levels; limitations	—

Appendix B — Screenshot Reference List

Capture each screenshot in your Kali lab and save it to `~/CS3/screenshots/` with the exact filename shown, then paste it into the matching figure placeholder in Sections 2–4.

Fig.	Filename	What to show
A0	figure_A0_evidence_hash.png	Both sha256sum results and the diff confirmation
A1	figure_A1_volatility_version.png	python2 vol.py --info banner (version)
A2	figure_A2_imageinfo.png	imageinfo: suggested profiles, DTB, KDBG, image time
B1	figure_B1_volatile_environment.png	Volatile Environment: USERNAME=IEUser, USERPROFILE=C:\Users\IEUser
B2	figure_B2_profilelist.png	ProfileList subkeys and the two SID …-1000 / …-1002 outputs
B3	figure_B3_pslist.png	pslist highlighting cmd.exe PID 3996 and conhost.exe PID 3988
B4	figure_B4_netscan.png	netscan: CLOSED 192.168.56.8 → 192.168.56.5:80 and sshd listener
B5	figure_B5_cmdscan.png	cmdscan: ipconfig, cd Downloads, copy secret_file.docx F:
B6	figure_B6_consoles.png	consoles: shell transcript including 1 file(s) copied.
B7	figure_B7_usbstor.png	USBSTOR key and the disk instance ID
B8	figure_B8_mounteddevices_F.png	MountedDevices: the \DosDevices\F: entry
B9	figure_B9_shellbags.png	shellbags: Downloads folder and the volume list with F:\

Appendix C — Volatility 2 Command Reference

Every command used in this report, in execution order. Replace the profile only if your own `imageinfo` suggests a different one.

```
# Profile identification
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem imageinfo

# Process analysis
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 pslist
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 psscan

# Network analysis
python2 ~/volatility/vol.py -f working/memdumpWin7 working.mem --profile=Win7SP1x86 23418 netscan

# Command history
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 cmdscan
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418
consoles

# Hive list
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418
hivelist

# User identification
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        printkey -K "Volatile Environment"
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList"
python2 ~/volatility/vol.py -f working/memdumpWin7 working.mem --profile=Win7SP1x86 23418 \
        printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-
909560446-1177810406-1000"
python2 ~/volatility/vol.py -f working/memdumpWin7 working.mem --profile=Win7SP1x86 23418 \
        printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-
909560446-1177810406-1002"

# USB and drive-mapping artefacts
python2 ~/volatility/vol.py -f working/memdumpWin7 working.mem --profile=Win7SP1x86 23418 \
        printkey -K "ControlSet001\Enum\USBSTOR"
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        printkey -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00"
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}"
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        printkey -K "MountedDevices"

# File Explorer history
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418
shellbags
python2 ~/volatility/vol.py -f working/memdumpWin7_working.mem --profile=Win7SP1x86_23418 \
        shellbags --output=html --output-file=reports/shellbags.html
```

Appendix D — Chain-of-Custody Worksheet

Field	Value
Case / lab identifier	SBT-DF204-CaseStudy3-Ibrahim-Diseh-Garba
Trainee name	Ibrahim Diseh Garba
Student ID	2025/FWSD/11521
Date and time acquired	4 October 2026 — 04:45 WAT
Evidence file(s)	memdumpWin7.mem (preserved); memdumpWin7_working.mem (analysis copy)
Source	ICDFA SBT-DF204 Case Study 3 lab package
File type	Windows 7 x86 (PAE) raw physical memory dump
File size	1,073,676,288 bytes
Original SHA-256	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
Original MD5	ac7eaa46a7006d5eb13cda13ee0b9bc9
Working-copy SHA-256	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
Storage location	~/CS3/evidence/ and ~/CS3/working/
Custodian	Ibrahim Diseh Garba (Student, ICDFA)
Handling notes	Original preserved read-only; analysis on working copy only; no live website contacted; no recovered file executed; no password or hash cracking performed; unrelated personal data redacted
Analysis tools	Volatility 2.6.1, python2 2.7.18, pycryptodome 3.23.0, distorm3 3.5.2, Kali Linux (rolling)
Analysis date	4 October 2026

Appendix E — Figure Index

Figure	Filename	Cited in
A0	figure_A0_evidence_hash.png	Section 2.3
A1	figure_A1_volatility_version.png	Section 2.4
A2	figure_A2_imageinfo.png	Section 3.2
B1	figure_B1_volatile_environment.png	Section 4 Q3.1
B2	figure_B2_profilelist.png	Section 4 Q3.2–3.5
B3	figure_B3_pslist.png	Section 4 Q4.1
B4	figure_B4_netscan.png	Section 4 Q4.2
B5	figure_B5_cmdscan.png	Section 4 Q5.1
B6	figure_B6_consoles.png	Section 4 Q5.2
B7	figure_B7_usbstor.png	Section 4 Q6.1–6.2
B8	figure_B8_mounteddevices_F.png	Section 4 Q6.4
B9	figure_B9_shellbags.png	Section 4 Q7